ARTICLE

Timing or tumour? The evidence behind glioma progression scores on post-treatment MRI

Sri Datta Ganesh Bandreddi[a] and Jessica Zhang[b]

[a]Department of Biomedical Engineering, Carnegie Mellon University, Pittsburgh, PA, USA; [b]Department of Mechanical Engineering, Carnegie Mellon University, Pittsburgh, PA, USA

ARTICLE HISTORY

Compiled November 1, 2026

ABSTRACT

A progression score from surveillance MRI after glioma treatment can rank scans largely by when they were taken, without showing which evidence carried it. PRISM splits each score into image and treatment-timeline evidence, attaches a confidence to each, and has a local language model write reports. At MU-Glioma-Post, the development centre, the timeline reached an AUC of 0.82, and the locked combined score reached 0.77 at LUMIERE. In exploratory analyses, within a phase of care at MU images ranked scans as well as the timeline (0.68 against 0.64), and at UCSF-ALPTDG the scores followed the enhancement on which progression was called. Pathway disagreement marked less reliable scores at MU only. After one prompt revision, all 406 reports passed an automatic check against their records. In these cohorts, what carried the score depended on when patients were scanned and how progression was labelled. No clinician has used the reports yet.

KEYWORDS

glioma surveillance; external validation; clinical decision support

1. Introduction

Glioblastoma and other aggressive gliomas are followed with repeated MRI after surgery and chemoradiotherapy, and each scan is read against earlier scans and the course of treatment. The call is hardest soon after radiotherapy, when treatment can increase contrast enhancement without tumour growth. This pseudoprogression is concentrated within 12 weeks of completing chemoradiotherapy (Wen et al. 2023), and is more common in MGMT promoter-methylated tumours (Taal et al. 2008; Brandes et al. 2008); months later, radiation necrosis can mimic progression in the same way.

A progression score from such scans can look better than it is, for three reasons. Scan timing follows treatment, so a model can rank scans by when they were taken rather than by what the tumour did. The hardest situations, such as growing enhancement soon after radiotherapy, are sparsely represented in labelled cohorts, so a model gives a number where it has seen few cases.

CONTACT Sri Datta Ganesh Bandreddi. Email: sbandred@andrew.cmu.edu; ORCID: 0009-0004-3617-6734

And centres hold different data and document progression differently, so a model fitted at one centre meets missing inputs and a different label at the next. Only 3 of 20 studies in a recent review of automated response assessment tested their models externally (Shi et al. 2025). A single probability shows none of this; a system can address each problem, with separate pathways that expose the schedule, a record of the situations with few cases, and modules that run on whatever a centre records.

Language models now write fluent clinical text, but a general-purpose chat model cannot answer the question a radiologist asks of a follow-up scan: does the signal come from the tumour or from the treatment schedule, and has the system been tested on scans like this one? It does not measure the tumour, does not know how a model performed in a cohort, may state numbers held in no record, and, if hosted outside the hospital, may not be permitted to receive patient data.

We built PRISM to make that question answerable at each scan. The name is not an acronym: as a prism splits white light into its colours, PRISM splits a progression score into the evidence behind it: an image pathway that uses only the segmented tumour and its change, a timeline pathway that uses clinical data and the position in treatment, and a combined score, each with a stated confidence. A language model running inside the hospital only puts the resulting record into words: it is given no images and no calculations to make, and every number it writes is checked against the record. We developed PRISM on MU-Glioma-Post (MU) and applied its models unchanged to LUMIERE and UCSF-ALPTDG (UCSF). Each part of PRISM has a precedent (Section 4); the paper makes four contributions:

(1) **A measurement of what carries the score.** Scan timing is known to carry much of a pseudoprogression model's output (Gomaa et al. 2025) and has been noted as cohort-level signal in LUMIERE (Tarek et al. 2026). We measure it: within phases of care, as a covariate-specific AUC (Janes and Pepe 2008), and across centres, where the same locked score was carried mostly by the treatment schedule at MU and by the enhancement on which progression was called at UCSF, so that there the score was partly graded against its own input (Worster and Carpenter 2008) (Section 3.2).

(2) **A per-scan confidence built from measured parts.** Separate image, clinical and combined models are common in oncology (Boehm et al. 2022), and disagreement between modalities (Mahapatra et al. 2026) and performance on similar cases (Habas et al. 2007) have each served as a measure of confidence. PRISM shows the image and timeline scores beside the combined score at every scan and states a confidence that joins a conformal set, the disagreement between the two pathways and the performance measured in that scan's phase of care and enhancement state (Section 2.1). Of these parts, average coverage held at all three centres on point estimates, and disagreement marked less reliable scores at MU only (Section 3.3).

(3) **Locked tests of a scan-level progression model at two external centres.** Under transfer rules fixed before the test data were opened, the combined model met its rule at LUMIERE and the image models met theirs at UCSF; the timeline did not transfer to UCSF (Section 3.1).

(4) **A check on every report.** As in earlier work in which local models wrote brain-tumour reports from extracted measurements (Heras Rivera et al. 2026; Valerio et al. 2025), the language model sees no images; PRISM adds a deterministic check of each report against its record (as for clinical summaries, Zeba et al. 2026). After one prompt revision every report passed; of seven local models given that prompt, which was revised for the model we used, only the two largest dense models passed in every cohort (Section 3.4).

2. Methods

2.1. PRISM

PRISM writes everything it computes for a scan, with the source of every value, into one structured record (Figure 1A; Table 1 names each score).

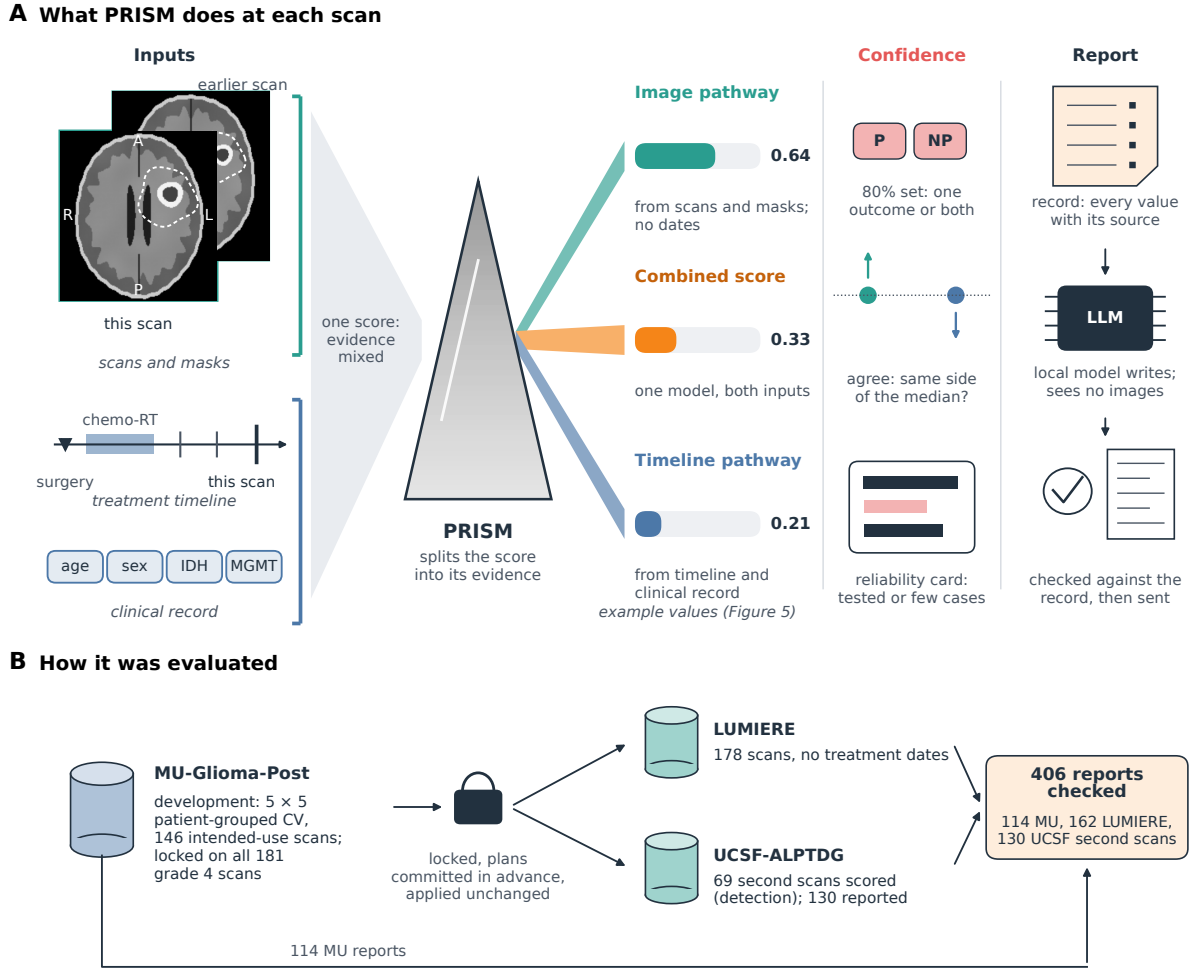


Figure 1. PRISM and how it was evaluated. (A) As a prism splits light, PRISM splits a progression score into its evidence; coloured brackets mark which inputs each pathway keeps. Measurements of the segmented tumour and its change feed the image pathway; clinical data and the position in treatment feed the timeline pathway; one model on both gives the combined score (values from the synthetic example of Figure 5). A three-part confidence is attached, every value enters one record with its source, and a local language model, given no images, writes the report, which a checker tests against the record. (B) Models were developed on MU by repeated cross-validation, then fitted once and applied unchanged at LUMIERE and UCSF under plans committed in advance.

The timeline pathway scores a scan from clinical data and the patient's position in treatment, both taken as of the scan day. Where a cohort lacks treatment dates, a reduced timeline keeps the inputs it has (Table 1), with the flag for the first weeks after radiotherapy taken from the rater's note at LUMIERE and approximated at UCSF. The image pathway scores the scan from 27 measurements of the segmented tumour and its change (Table 1) and uses no dates or intervals. LUMIERE's masks

Table 1. The scores built from measurements and records. Every score listed is a ridge logistic regression except enhancing volume alone, which is not fitted; the others are fitted within the training data, with unknown IDH or MGMT status as the reference category and missing measurements imputed with an indicator. The fine-tuned networks are described in Section 2.3. Status: planned before the analysis, or not fixed in advance (†).

Score	Inputs	Dates	Status
Clinical data alone	Age, sex, IDH and MGMT status	No	Planned reference (LUMIERE †)
Timeline pathway	Clinical inputs; days since diagnosis and surgery; radiotherapy start, end and dose; chemotherapy; visit index	Yes	Planned
Reduced timeline	Clinical inputs; days since surgery; flag for the first weeks after radiotherapy	Partly	Planned at UCSF-ALPTDG; † at LUMIERE
Image pathway	27 measurements of the segmented tumour: volumes, shape and intensity, change since the previous scan, volumes relative to the patient's smallest	No	† on MU; planned at UCSF-ALPTDG and in the system tests
Single-scan images	The 19 measurements of the current scan	No	† on MU; planned at UCSF-ALPTDG
Combined score	Inputs of both pathways; externally, the reduced timeline, volumes and their change, and the interval since a previous scan	Yes; partly externally	† on MU (full form); planned and locked (reduced form) at LUMIERE and UCSF-ALPTDG
Planned imaging model	Timeline inputs, image measurements and three schedule inputs (interval since the previous scan, whether there was one, smallest volume since the end of radiotherapy)	Yes	Planned primary model on MU; not part of PRISM
Enhancing volume alone	Log enhancing tumour volume, no fitted model	No	† reference

have three compartments and no separate cavity, so there the image pathway uses the three volumes and their change since the previous scan, a reduction fixed before evaluation. The measurements come from the masks released with each cohort (Mahmoud et al. 2025; Suter et al. 2022; Fields et al. 2024).

The confidence has three parts. A patient-level split conformal set at the 80% level (Angelopoulos and Bates 2023) holds one outcome when the score is clear and both when it is not. It comes from the combined model refitted on 70% of the MU patients and calibrated on the other 30% (within each training fold on MU) and is applied unchanged elsewhere, which assumes that external scans are exchangeable with MU's. The two pathways agree when both scores lie on the same side of their medians over the cohort's evaluation scans; because cohort medians need a batch of scans, we also tested medians fixed on MU, which can be applied to a single scan. The reliability card† (Mitchell et al. 2019) describes the scan's situation, its phase of care crossed with the change in enhancing volume, from that cohort's held-out results. Because those results include the scan itself, the card describes it slightly optimistically. The card gives AUCs for a situation with at least 10 scans of each outcome (tested), counts only for one with fewer (few cases), and marks one with none as not tested.

A local language model (Qwen3.8-27B, temperature 0) writes the report from the record alone, so patient data do not leave the hospital. A deterministic checker rejects any number, flag or confidence statement that is not in the record, any missing required content, and any wording that offers a diagnosis, advice or a probability (Supplementary Section S19).

2.2. Cohorts and the task

Grade 4 glioma is the primary population in all three cohorts (Table 2); we fitted each model once on MU and applied it unchanged, or locked, to the others. The task is to decide whether each scan is the one at which the cohort documented progression. On MU a positive is the first scan on or after the documented progression date, and a negative is a scan followed by at least 120 days without progression. At LUMIERE a positive is the first scan rated as progressive disease, and a negative is an earlier scan rated stable or responding. UCSF holds one pair of consecutive scans per patient, and we evaluated the second: a positive is a second scan whose progression date lies within 30 days of it and closer to it than to the first scan, and a negative is followed by at least 120 days without progression. On MU the intended use is scans from the start of radiotherapy, divided into three phases of care: during radiotherapy, early after it (up to 12 weeks after its end) and later; MU results are for the intended use unless stated. Predicting progression within 120 days of a scan is a secondary task.

2.3. Evaluation

The locked models were fitted on all 181 grade 4 detection scans of MU, from every phase of care, and the external tests followed plans committed before any score was compared with an external label. One deviation: the plans named all MU detection scans and we fitted on the grade 4 ones; fitting on all 229 scans of gliomas of every grade changed the LUMIERE AUC by less than 0.01. On MU we evaluated every model by five repeats of five-fold cross-validation grouped by patient and report the mean pooled held-out AUC with a 95% confidence interval (CI) from 1,000 patient-level bootstrap draws; external AUCs carry a 95% CI from the same patient-level bootstrap. The MU intervals resample held-out predictions and do not include the variance from refitting. No patient contributed to more than one of fitting, calibration and testing. We used repeated splits because single splits are unstable in small cohorts (An et al. 2021; Varoquaux 2018). An earlier radiomics model on MU, for example, reached 0.80 on one test split but averaged 0.72 when re-selected and re-tested over 500 splits of the same design (Supplementary Section S8 and Figure S2). As a reference, we also fitted each UCSF model on UCSF by cross-validation. To test whether deep networks see more than the measurements, we fine-tuned segmentation networks to score scans directly and compared them with the same networks trained on permuted labels. Table 3 lists the questions fixed in advance and their answers. Results of analyses not fixed in advance carry a dagger (†). The disagreement flag and its two tests were planned, but the median rule that defines disagreement was written down after the results had been seen, so every disagreement result carries a dagger.

3. Results

3.1. The timeline led at MU, and the combined and image models transferred without refitting

In the intended use on MU (146 scans), the timeline pathway detected the progression scan with an AUC of 0.82 (95% CI 0.75–0.90) and the image pathway with 0.71 (0.63–0.78)†; clinical data alone scored 0.53 (0.45–0.61; Figure 2). The combined score also reached 0.82 (0.74–0.89)†, with

Table 2. The three cohorts. What each cohort records, and the scans in the detection task. Grade 4 glioma is the primary population.

	MU-Glioma-Post	LUMIERE	UCSF-ALPTDG
Role	Development (repeated cross-validation)	External, locked	External, locked
Access	Public	Public	Data use agreement
Scans per patient	Up to 6 scans with a 120-day label (median 2)	Up to 10 scans with a 120-day label (median 2)	One consecutive pair
Treatment dates	Surgery, radiotherapy start and end, dose, chemotherapy	Days from surgery only; rater flag for scans within three months of radiotherapy	Days from surgery; radiotherapy start only
Documented progression (the label)	Documented date in the clinical record	Expert response rating of each scan	Progression date supplied with the dataset
Segmentation masks	Automatic masks released with the dataset	DeepBraTumIA (three labels)	Masks provided with the dataset
Patients[a]	126	61	140
Age, median	62.5 (IQR 53–70)	62 (IQR 53–70)	57.5
Male	66 (52%)	32 (52%)	81 (58%)
IDH wildtype / mutant / unknown	112 / 8 / 6	38 / 1 / 22	81 / 13 / 46
MGMT methylated / unmethylated / unknown	45 / 62 / 19	26 / 28 / 7	71 / 43 / 26
Detection task[b]: progression / non-progression scans (patients)	62 / 84 (80) in the intended use; 64 / 117 (92) over all scans	73 / 105 (79)	29 / 40 (69), second scan only
Phase of care in the detection set	Progression: 56 of 62 later after radiotherapy; non-progression: 51 of 84 early after radiotherapy	Not recorded	65 of 69 later after radiotherapy[c]

[a] Patient characteristics describe the cohorts of the secondary 120-day task at MU-Glioma-Post (grade 4) and LUMIERE, and all grade 4 patients at UCSF-ALPTDG. Age is at diagnosis at MU-Glioma-Post and at surgery at LUMIERE.

[b] Positives are the scan at which progression was documented; negatives are defined per cohort in Section 2.2. The LUMIERE detection set has more patients than the 120-day cohort because it includes patients who contribute only a progression scan. At MU-Glioma-Post, the 146 scans of the intended use are the 181 grade 4 detection scans less the 35 taken before radiotherapy started, and 42 grade 4 progression scans of patients with no scan carrying a 120-day label are excluded. Reports were evaluated on the 114 intended-use scans of the 60 MU-Glioma-Post patients not used to develop the prompt, on 162 of the 178 LUMIERE scans and on 130 of the 140 grade 4 second scans at UCSF-ALPTDG, the rest belonging to development patients (Section 3.4).

[c] UCSF-ALPTDG gives only the first radiotherapy start; the phase assumes a six-week course. Two scans were early after radiotherapy and two had no radiotherapy date.

Table 3. Tests fixed in advance. Each rule was committed to the repository before the analysis it governs. AUC differences with 95% CIs.

Question	Rule fixed in advance	Outcome
Detection on MU-Glioma-Post		
Do image measurements add to the timeline?	Planned model ≥ 0.75 (lower bound ≥ 0.65); gain ≥ 0.03, CI above 0	**No**: AUC 0.837 (0.763–0.904); gain +0.015 (−0.032 to +0.063)
Do fine-tuned deep networks add?	Gain ≥ 0.03, CI above 0; permuted-label control covers 0.5	**No**: no network gained; control valid, 0.483 (0.436–0.527)
External cohorts (LUMIERE, UCSF-ALPTDG)		
Combined score at LUMIERE	AUC ≥ 0.65	**Yes** (Section 3.1)
Image pathway at UCSF-ALPTDG	AUC ≥ 0.65, lower bound > 0.5	**Yes** (Section 3.1)
Locked image pathway adds to locked clinical data at UCSF-ALPTDG	Gain, CI above 0	**Yes**: +0.248 (+0.096 to +0.392)
Deep networks learn at UCSF-ALPTDG	Above the permuted-label control; control covers 0.5	**Yes**: +0.201 (+0.094 to +0.306) over a control of 0.541 (0.480–0.603)
The system (MU-Glioma-Post, LUMIERE, UCSF-ALPTDG)		
Conformal coverage	≥ 0.78 at the 80% level in each cohort	**Yes** for the combined score, on point estimates; not for images at LUMIERE (0.708) or the timeline at UCSF-ALPTDG (0.696) (Section 3.3)
Does the combined score rank flagged scans less well?	AUC unflagged minus flagged, CI above 0	Disagreement†: **MU only**. Set holding both outcomes: **MU and LUMIERE**[a]
Does setting flagged scans aside beat matched removal?	Retained minus matched-removal AUC, CI above 0	Disagreement†: **MU only**. Set holding both outcomes: **UCSF only**[a]
Do reports pass the lexical check?	≥ 95% in each cohort	First evaluation **No** (every LUMIERE report failed); after one revision **Yes** (Section 3.4)
Is the tier stated where evidence is thin?	100%	**Yes** (Section 3.4)
Do reports follow a changed record?	≥ 27 of 30 perturbed records	**Yes** (Section 3.4)

Plans: detection (commit 2b4c927), UCSF-ALPTDG (0e7c9fe), the system (e7cba80), second report evaluation (4f7e7b9). Matched removal keeps the phase-of-care mix of the removed scans (random removal at LUMIERE, which has no phase). Over all scans of gliomas of all grades (a planned secondary population), the planned model gained +0.041 (+0.006 to +0.081) over the timeline, which may reflect its schedule inputs (Section 3.2).

[a] With each cohort's own medians, the rule the reports use; the median rule was written down after the results had been seen†. With medians fixed on MU, disagreement also met both tests at LUMIERE.

no detectable gain over the timeline (−0.001, −0.05 to +0.05). A planned model that added image measurements and three schedule inputs to the timeline also fell short of its target gain of 0.03 (Table 3). Within phases of care the timeline's gain was not detectable†, although its interval cannot exclude a gain as large as the overall one (Section 3.2). Forecasting progression within 120 days was harder, at 0.60 (0.52–0.69) in the intended use (Supplementary Section S23).

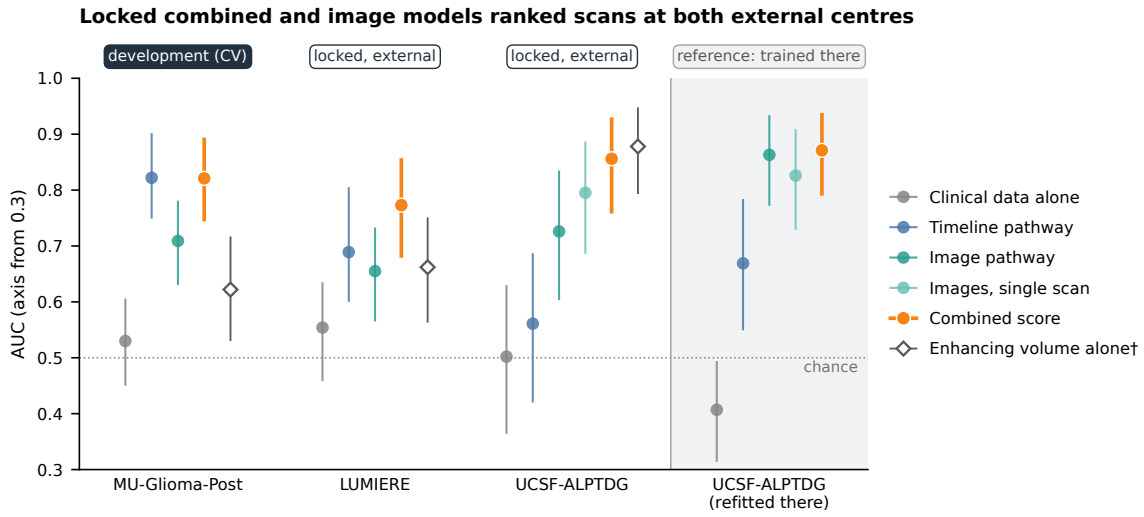


Figure 2. The combined and image models rank scans at two external centres without refitting. AUC for detecting the scan at which progression was documented. MU: repeated cross-validation, intended use. LUMIERE and UCSF: models fitted once on MU and applied unchanged, with the reduced timeline. UCSF refitted there: reference, where clinical data alone fall below 0.5, consistent with the downward bias of cross-validated AUC without signal. At MU the combined score is its full form; externally, its reduced form. At UCSF the images are shown for the single-scan model and for the image pathway. The image pathway and combined score on MU and the clinical-only, reduced-timeline and image models at LUMIERE were not fixed in advance†. Open diamond: log enhancing tumour volume alone† (at LUMIERE, the 163 scans with enhancing tumour). Points with 95% patient-bootstrap CIs on an axis from 0.3; dotted line, chance.

The locked combined model, in the reduced form used externally (planned; 0.80, 0.73–0.89, on MU by cross-validation), reached 0.77 (0.68–0.86) at LUMIERE and met its planned transfer rule (Table 3); its gain over the reduced timeline† was 0.08 (−0.01 to +0.16). At UCSF both planned image models transferred, the single-scan measurements reaching 0.80 (0.69–0.89) and the image pathway 0.73 (0.60–0.84); with coefficients fitted on MU, the image pathway scored 0.07 below the single-scan model (−0.13 to −0.01), and retrained at UCSF it gained 0.14 (0.03–0.26) over its locked version. The reduced timeline scored 0.56 (0.42–0.69) at UCSF, where its radiotherapy flag is approximated. The locked combined score reached 0.86 (0.76–0.93), but part of this reflects scan timing: the interval between the two scans was shorter before progression†.

As a probability, the combined score showed no detectable miscalibration in the large at LUMIERE† (observed over expected 0.97, 0.81–1.14) but overpredicted progression at UCSF (0.72, 0.55–0.91), and a threshold at the MU median† called 90% of UCSF scans positive.

3.2. The scores followed the treatment schedule at MU and enhancement at UCSF

At MU the timeline and combined scores depended mainly on where the patient was in treatment, and at UCSF the image scores followed the volume of enhancing tissue† (Figure 3). At MU most

progression scans came more than 12 weeks after radiotherapy and most non-progression scans within those 12 weeks (Table 2), so the schedule alone could separate most of them. A within-phase AUC pools only the pairs of a progression and a non-progression scan from the same phase of care; most such pairs come from the later phase (56 progression and 32 non-progression scans). Adding the timeline to clinical data and images† changed the AUC by +0.11 (0.04–0.20) across phases of care and by −0.02 (−0.13 to +0.11) within them (Figure 3A). Within a phase, the image pathway scored 0.68 (0.58–0.77) and the timeline 0.64 (0.51–0.78), with no difference detectable.

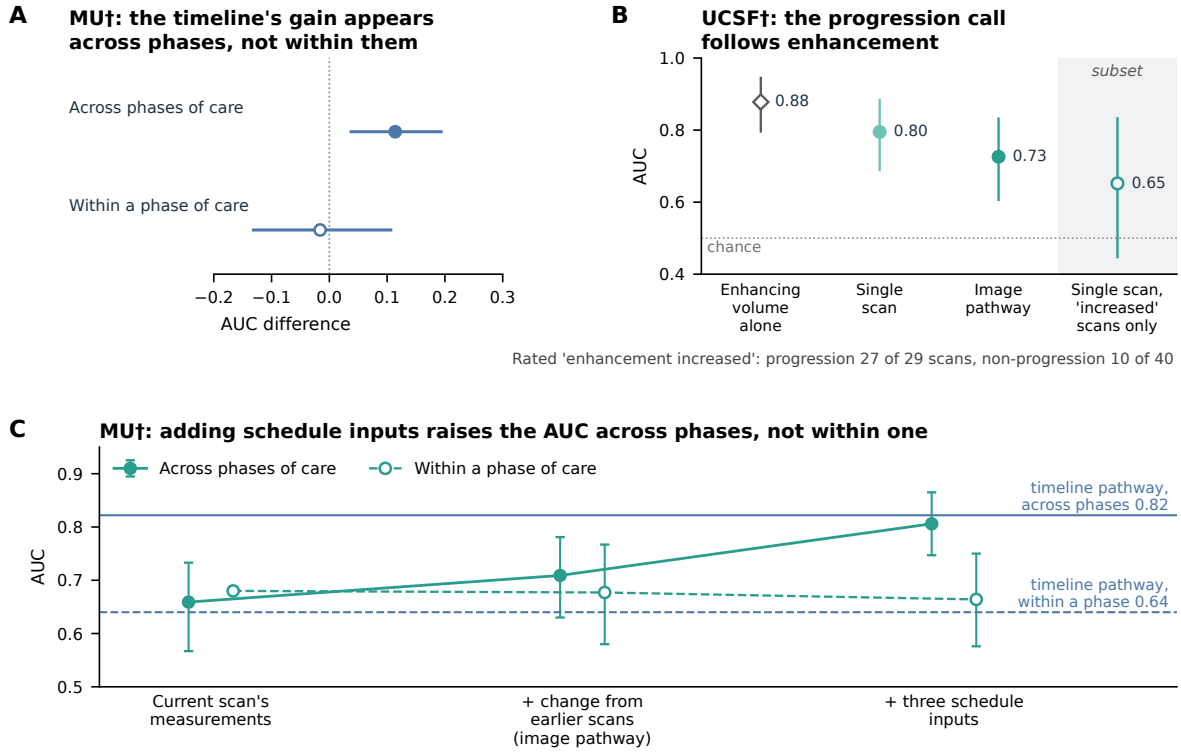


Figure 3. The scores followed the schedule at MU and enhancement at UCSF. (A) MU: gain from adding the timeline to clinical data and images, across and within phases of care. (B) UCSF: enhancing volume alone beside the locked image models; the open point is restricted to the 37 scans rated as showing increased enhancement. (C) MU: the image model built in three steps, across and within phases of care, against the timeline pathway; the third step adds the planned model's schedule inputs to the image inputs, without its timeline. The gain at the second step has an interval that includes zero, and the first within-phase point has no interval. The audits were not fixed in advance (†); the locked image models' AUCs over all 69 scans in (B) were planned. Points with 95% patient-bootstrap CIs.

On MU the image model gained most from inputs that carry timing†. Adding change from earlier scans to the current scan's measurements, which gives the image pathway, raised the AUC by 0.05, with an interval that included zero (−0.003 to +0.106). Adding the planned model's three schedule inputs (Table 1) raised it by a further 0.10 (0.05–0.15) across phases of care but not within them (−0.01, −0.06 to +0.04) (Figure 3C).

At UCSF the progression labels coincided with a rated increase in enhancement†, so a model could score well there by tracking it: UCSF rated enhancement as increased at 27 of the 29 progression scans, and progression was dated a median of 7 days after the scan (Figure 3B). Enhancing volume alone reached 0.88 (0.79–0.95) there, above the point estimates of the locked image mod-

els, and above its AUC at MU and LUMIERE (Figure 2). Among the 37 scans rated as increased, the single-scan model reached 0.65 (0.44–0.84), so this sample cannot show whether the measurements add to that rating. Deep networks fine-tuned on UCSF scans beat the permuted-label control (Table 3), and their scores tracked enhancing volume†.

3.3. Coverage meets its target on point estimates; disagreement marks scans the combined score ranks less well at MU

The 80% conformal sets of the combined score met the planned coverage threshold of 78% on point estimates in all three cohorts, covering 78% to 87% of scans, and held a single outcome for 62% to 75% of scans (Figure 4A); the intervals at MU (76% to 88%) and UCSF (68% to 87%) reached below the threshold, and the image set at LUMIERE and the timeline set at UCSF fell below it. Coverage differed by outcome at the external centres†. LUMIERE covered 77% (67% to 87%) of progression scans and 94% of non-progression scans, and UCSF 97% of progression and 65% (51% to 79%) of non-progression scans. Calibrating the sets separately for each outcome on MU did not fix this, and non-progression coverage fell to 73% (63% to 82%) even on MU.

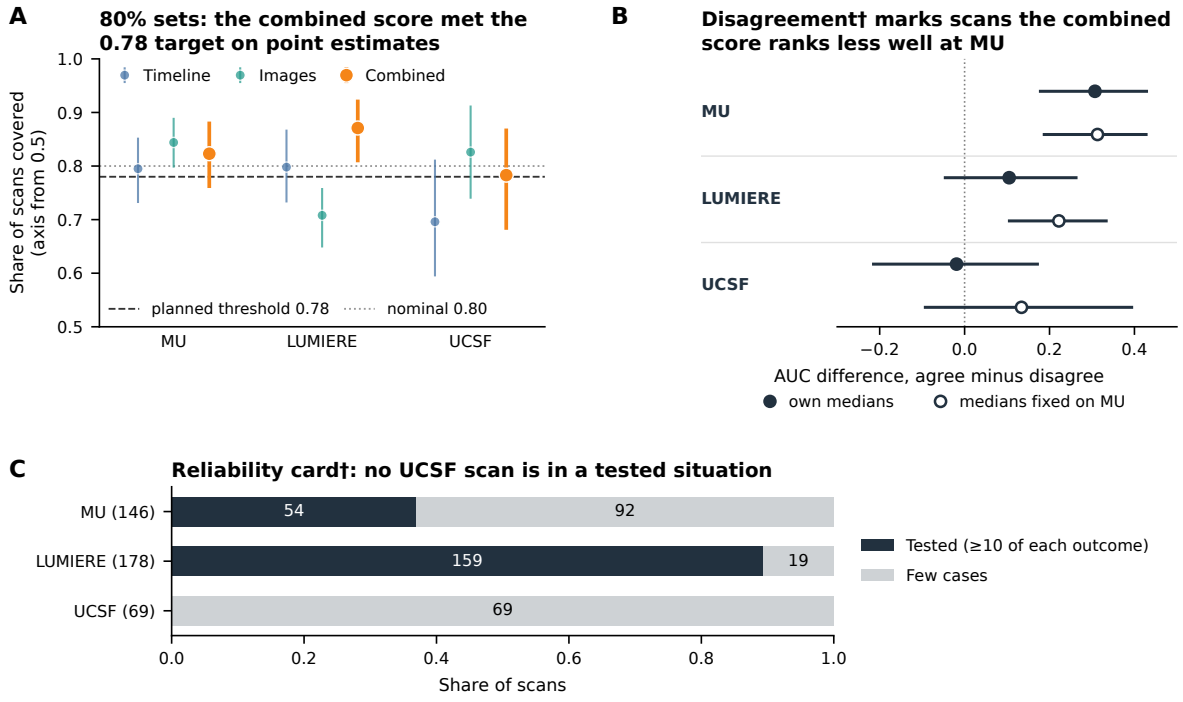


Figure 4. The combined score's conformal sets meet their coverage target on point estimates; disagreement† marks scans the combined score ranks less well at MU. (A) Coverage of the 80% conformal sets by cohort and pathway; dashed line, the planned threshold (0.78); dotted line, the nominal level (0.80). (B) The combined score's AUC where the pathways agree minus where they disagree, with each cohort's own medians (filled) and with medians fixed on MU (open; on MU, the other folds' medians). The tests were planned; the median rules were not†. (C) Scans whose situation reached the reliability card's† tested tier; the rest had few cases, and none was untested.

Disagreement between the pathways† marked scans that the combined score ranked less well at MU, but not at the external centres. On MU the pathways disagreed on 40% of scans; the com-

bined score's AUC was 0.31 higher where they agreed (0.18–0.43), and setting the disagreeing scans aside gained 0.08 (0.05–0.12) over removing as many scans matched for phase of care, meeting both planned tests (Table 3; Figure 4B). With each cohort's own medians, the rule the reports use, neither test was met at LUMIERE or UCSF; with medians fixed on MU, both were met at LUMIERE but not at UCSF (Figure 4B). A conformal set holding both outcomes, used as a flag, met one of the two planned tests in each cohort (Table 3).

Only four situations on the reliability card† reached the tested tier (Figure 4C), one at MU and three at LUMIERE, where no phase of care is recorded and situations differ by enhancement only; no scan in these sets was in an untested situation. Every UCSF situation had few cases, and at MU no situation in the first 12 weeks after radiotherapy held more than three progression scans. In the planned descriptive comparison, tested situations had lower error than few-case ones at LUMIERE (29% against 42% of scans misclassified at the cohort median) but not at MU (29% against 24%), where the few-case scans are mostly clear non-progression early after radiotherapy.

3.4. After one revision, every report passed an automatic check against its record

Every report written by Qwen3.8-27B passed the lexical check against its record, 114 at MU, 162 at LUMIERE and 130 at UCSF; of six other local models given the same prompt, which had been revised for Qwen3.8-27B, only Gemma-4 31B also did (Figure 5B). All 220 reports in few-case or untested situations said so (71 at MU, 19 at LUMIERE and 130 at UCSF). When one confidence or measurement field was changed in 30 records from the three cohorts, 28 reports followed the change, against a planned minimum of 27 (Table 3). These results followed one revision of the prompt†. Before it, every LUMIERE report failed because a fixed sentence quoted a number held in no field, so we corrected the prompt, left the checker unchanged and evaluated again on patients not used for the revision. The check confirms that a report matches its record; it cannot judge emphasis, what a clinician would infer, or whether the segmentation is right.

4. Discussion

PRISM's combined and image models ranked scans at two other centres without refitting, but the evidence behind a score depended on how each cohort was scanned and labelled: mostly the treatment schedule at MU and, at UCSF, the enhancement on which progression was called†. An AUC cannot show which evidence a model used; the audits of Section 3.2 suggested it, and the separate pathways show the clinician which kind of evidence drives each score. Process variables can carry much of the apparent skill of an image model (Badgeley et al. 2019), and image models for glioblastoma follow-up have been benchmarked separately at the first and second scans after radiotherapy (Guo and Mirzaei 2025); we found no earlier study that compared image and timeline evidence within phases of care, or showed that what carries the same locked score differs between centres tested externally. The risk scale did not transfer to UCSF, so a new centre would recalibrate the score and set any threshold on local scans.

Because its modules are separate, PRISM runs on what a cohort holds, given scans and masks. The image pathway needs only those, which most public datasets have, and a single-scan version serves where there is no earlier scan. The timeline is added where dates exist. The reliability card

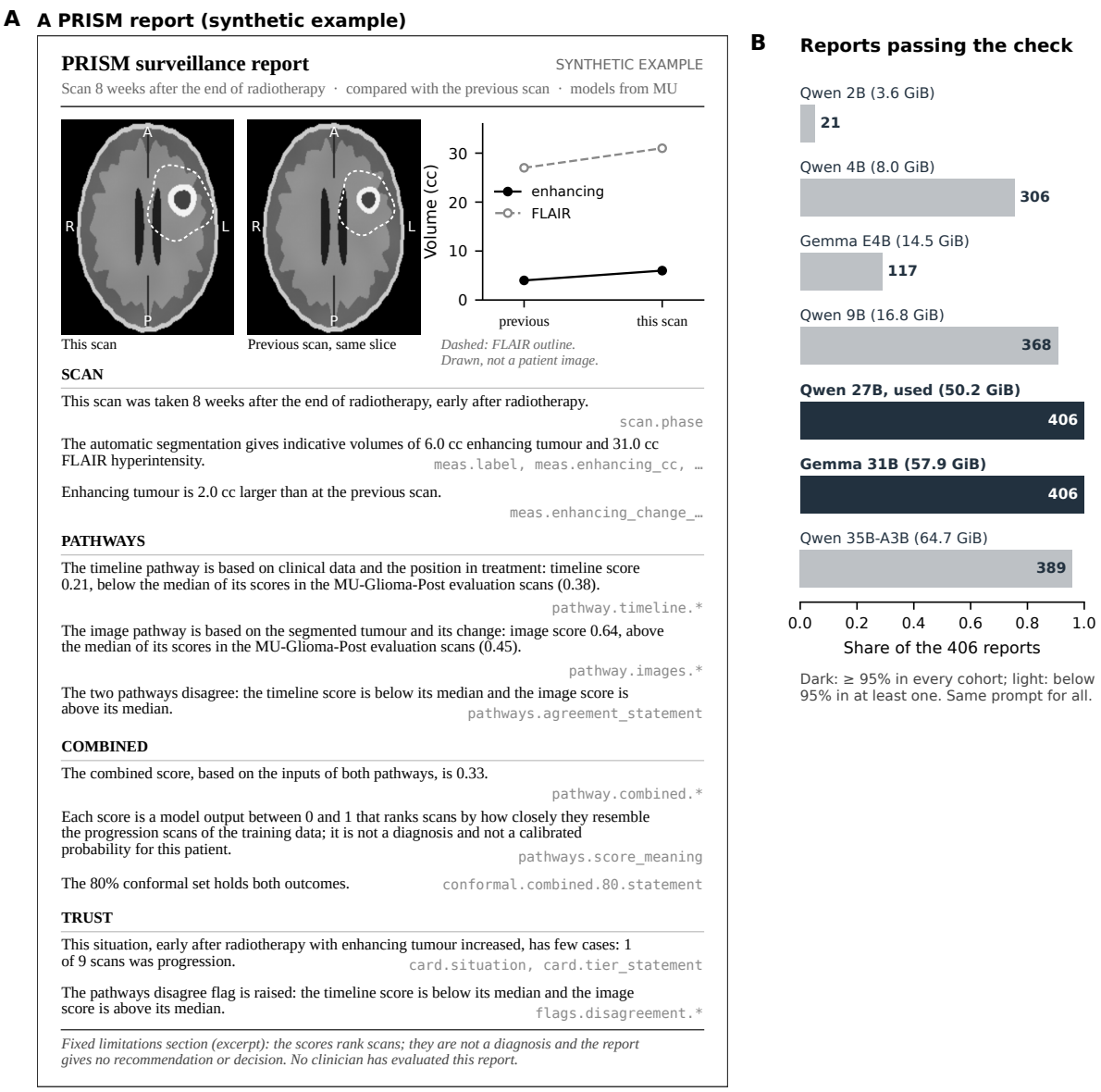


Figure 5. What the clinician receives, and the check. (A) A report for a synthetic scan eight weeks after radiotherapy with growing enhancement (Section 4): the segmentation figure, drawn here rather than taken from a patient, and the narrative. Volumes and scores are invented; the card row is MU's. Each sentence lists, in grey, the record fields it cites. The delivered report also holds tables of every value and a fixed limitations section (excerpt shown). (B) Reports passing the check, of 406, for seven local language models given the same prompt (Qwen3.5 2B, 4B and 9B; Gemma-4 E4B and 31B; Qwen3.8-27B, used in PRISM; Qwen3.5-35B-A3B), after one revision of the prompt† (Section 3.4); dark bars, at least 95% passed in every cohort, and the model weights' size is in brackets.

is a lookup table, so a new cohort adds rows to it and a situation's tier can change as evidence accumulates.

Consider a scan eight weeks after radiotherapy in which the enhancing tumour has grown (Figure 5A). The timeline pathway scores such a scan low, because most scans at this point in treatment are not progression, and the image pathway scores it high, because enhancement has grown. The report shows both, states that they disagree and that the conformal set holds both outcomes, and records from the card that the situation has few cases (at MU, 1 of 9 such scans was progression†). The evidence PRISM holds for this situation is therefore too thin to support a call, in the weeks when pseudoprogression is most common, and the clinician decides with information it does not use, such as symptoms, steroid dose and the radiotherapy plan.

Each component of PRISM has been described before: longitudinal segmentation with measurement of change (Rudie et al. 2022), late fusion of separately trained image and clinical models (Boehm et al. 2022), fusion of clinical and imaging data to separate pseudoprogression from progression (Gomaa et al. 2025), conformal prediction for glioma segmentation (Elyassirad et al. 2025) and for triage at a deployment site (Angelopoulos et al. 2024), disagreement between modalities as uncertainty (Mahapatra et al. 2026), reliability from performance on similar cases (Habas et al. 2007), model cards (Mitchell et al. 2019), and reports written by local language models from brain-tumour measurements (Heras Rivera et al. 2026; Valerio et al. 2025). The nearest systems combine some of these. One assigns a BT-RADS follow-up category from tumour volumes and treatment variables extracted by a language model, at one centre and without a per-scan uncertainty (Jabal et al. 2026); another keeps a ledger of evidence with a confidence for each claim in longitudinal neuro-oncology, but works from reports rather than images and checks its evidence with a language model (Liu et al. 2026). On these cohorts, earlier work predicted a patient-level label on MU (Christodoulou et al. 2026) and response categories on LUMIERE ([cite Tikhonov 2025, arXiv:2509.06511; full reference not yet verified]), and a model trained on MU was applied unchanged at UCSF to predict survival (Ding et al. 2026); separately, Chen et al. (2026) predicted recurrence risk from preoperative and postoperative MRI across centres. None reports image and timeline evidence separately at each scan, or tests a scan-level progression call locked at two external centres. What PRISM adds is therefore not a new architecture but a combination: image and timeline evidence shown at each scan with a confidence from measured performance, tested at two external centres under rules fixed in advance.

PRISM was developed at one centre, and the two external tests are its only evidence that the ranking transfers. The label is progression as each centre documented it; no scan was re-read, and at UCSF most progression scans were also rated as showing increased enhancement. Of the confidence measures, only average coverage met its planned threshold at all three centres, and only on point estimates. Disagreement† marked scans the combined score ranked less well at MU alone; that rule needs a batch of a cohort's scans, and the single-scan version (MU medians) was added after internal review. No UCSF scan fell in a tested situation. Subgroups are small, the analyses marked † need testing on new data, and no clinician has yet read the reports. PRISM forecast progression within 120 days less well than it detected progression at the scan (AUC 0.60, 0.52–0.69, in the intended use).

The next steps are to pool the rare situations (early progression after radiotherapy, and growing enhancement without progression) across cohorts, including the Burdenko and UCSD-PTGBM datasets, and test models fine-tuned on them there; and a reader study to show whether separate

evidence and a stated confidence change clinicians' decisions.

Conclusion

PRISM keeps image and treatment-timeline evidence apart and shows the clinician both scores at every scan with how far each has been tested there. Its combined and image models ranked scans at two other centres without refitting, while the scores followed the treatment schedule at MU and enhancement at UCSF, a difference the separate pathways make visible.

Acknowledgements

[PENDING AUTHOR DECISION].

Disclosure statement

[PENDING AUTHOR DECISION: competing interests, or "The authors report there are no competing interests to declare."] If any author serves on the editorial or advisory board of this journal, state it here.

Declaration of generative AI use

[DRAFT, PENDING AUTHOR CONFIRMATION:] Generative AI is part of the system studied. PRISM's report generator uses a local language model (Qwen3.8-27B; six other open-weight models in a comparison); all ran offline on the authors' computing cluster and saw only derived system records. An AI assistant (Claude, Anthropic) was used to write and run analysis code and to draft and edit the manuscript under the authors' direction; it was given no patient-level data through any hosted service. The authors reviewed all code, results and text and take full responsibility for the content of this article.

Funding

No funding was obtained for this work.

Ethics statement

[PENDING AUTHOR DECISION: secondary analysis of de-identified datasets (MU-Glioma-Post and LUMIERE, public; UCSF-ALPTDG under a data use agreement); state the institutional determination or waiver and cite the source datasets' approvals.] No registered protocol. Analysis plans were committed to the repository before each external test; their commits are listed in Table 3. Patient and public involvement: none.

Data availability statement

[PENDING AUTHOR DECISION: MU-Glioma-Post (TCIA, doi:10.7937/7K9K-3C83) and LUMIERE (figshare; Suter et al. 2022) are public; UCSF-ALPTDG is available from its provider under a data use agreement; state which derived tables, fold assignments and locked model files are released, with a DOI.] Code: [PENDING: public repository link and release tag.]

Supplemental material

Supplemental data for this article are provided as a separate file.

Notes on contributors

[PENDING: a short biographical note for each author (no more than 200 words each).]

References

An C et al. 2021. Radiomics machine learning study with a small sample size: Single random training-test set split may lead to unreliable results. PLOS ONE. 16(8):e0256152. https://doi.org/10.1371/journal.pone.0256152

Angelopoulos AN, Bates S. 2023. Conformal prediction: A gentle introduction. Found Trends Mach Learn. 16(4):494–591. https://doi.org/10.1561/2200000101

Angelopoulos AN et al. 2024. Conformal triage for medical imaging AI deployment. [Preprint, medRxiv]

Badgeley MA, Zech JR, Oakden-Rayner L et al. 2019. Deep learning predicts hip fracture using confounding patient and healthcare variables. npj Digit Med. 2:31. https://doi.org/10.1038/s41746-019-0105-1

Boehm KM, Aherne EA, Ellenson L, Nikolovski I et al. 2022. Multimodal data integration using machine learning improves risk stratification of high-grade serous ovarian cancer. Nat Cancer. 3:723–733. https://doi.org/10.1038/s43018-022-00388-9

Brandes AA, Franceschi E, Tosoni A et al. 2008. MGMT promoter methylation status can predict the incidence and outcome of pseudoprogression after concomitant radiochemotherapy in newly diagnosed glioblastoma patients. J Clin Oncol. 26(13):2192–2197. https://doi.org/10.1200/JCO.2007.14.8163

Chen C, Guo F, Huang D et al. 2026. Predicting recurrence risk of glioblastoma based on preoperative-postoperative longitudinal MRI: A multicenter study. Bioengineering. 13(6):668. https://doi.org/10.3390/bioengineering13060668

Christodoulou RC, Vamvouras G, Pitsillos R, Solomou EE, Georgiou MF. 2026. Explainable radiomics with probability calibration for postoperative glioblastoma surveillance. Eur J Radiol Artif Intell. 5:100074. https://doi.org/10.1016/j.ejrai.2026.100074

Ding T, Zou Y, Chen C, Shah M, Tian Y. 2026. CLARITY: Medical world model for guiding treatment decisions by simulating context-aware disease trajectories. [Preprint, arXiv:2512.08029]. https://arxiv.org/abs/2512.08029

Elyassirad D, Gheiji B, Vatanparast M, Ahmadzadeh AM, Faghani S. 2025. CONSeg: Voxelwise uncertainty quantification for glioma segmentation using conformal prediction. Am J Neuroradiol. 46(12):2553–2560. https://doi.org/10.3174/ajnr.A8914

Fields BKK et al. 2024. The University of California San Francisco adult longitudinal post-treatment diffuse glioma MRI dataset. Radiol Artif Intell. 6(4):e230182. https://doi.org/10.1148/ryai.230182

Gomaa A, Huang Y, Stephan P et al. 2025. A self-supervised multimodal deep learning approach to differentiate post-radiotherapy progression from pseudoprogression in glioblastoma. Sci Reports. 15:17133. https://doi.org/10.1038/s41598-025-02026-7

Guo W, Mirzaei G. 2025. Timepoint-specific benchmarking of deep learning models for glioblastoma follow-up MRI. Cancers. 18(1):36. https://doi.org/10.3390/cancers18010036

Habas PA, Zurada JM, Elmaghraby AS, Tourassi GD. 2007. Reliability analysis framework for computer-assisted medical decision systems. Med Phys. 34(2):763–772. https://doi.org/10.1118/1.2432409

Heras Rivera JE et al. 2026. BTReport: A framework for brain tumor radiology report generation with clinically relevant features. [Preprint, arXiv:2602.16006]. https://arxiv.org/abs/2602.16006

Jabal MS et al. 2026. Agentic automation of BT-RADS scoring: End-to-end multi-agent system for standardized brain tumor follow-up assessment. [Preprint, arXiv:2603.21494]. https://arxiv.org/abs/2603.21494

Janes H, Pepe MS. 2008. Adjusting for covariates in studies of diagnostic, screening, or prognostic markers: an old concept in a new setting. Am J Epidemiol. 168(1):89–97. https://doi.org/10.1093/aje/kwn099

Liu Y et al. 2026. TumorBoard: Evidence-grounded multi-agent decision support for longitudinal neuro-oncology. [Preprint, arXiv:2608.03190]. https://arxiv.org/abs/2608.03190

Mahapatra D et al. 2026. EnTrust: Modeling inter-modal conflict for trustworthy multimodal medical image analysis. [Preprint, arXiv:2606.21384]. https://arxiv.org/abs/2606.21384

Mahmoud E, Gass J, Dhemesh Y et al. 2025. MU-Glioma Post: A comprehensive dataset of automated MR multi-sequence segmentation and clinical features. Sci Data. 12(1):1847. https://doi.org/10.1038/s41597-025-06011-7

Mitchell M, Wu S, Zaldivar A et al. 2019. Model cards for model reporting. In: Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT* '19). ACM. p 220–229

Rudie JD, Calabrese E, Saluja R et al. 2022. Longitudinal assessment of posttreatment diffuse glioma tissue volumes with three-dimensional convolutional neural networks. Radiol Artif Intell. 4(5):e210243. https://doi.org/10.1148/ryai.210243

Shi T, Kujawa A, Linares C, Vercauteren T, Booth TC. 2025. Automated longitudinal treatment response assessment of brain tumors: A systematic review. Neuro-Oncology. 27(8):1946–1971. https://doi.org/10.1093/neuonc/noaf037

Suter Y, Knecht U, Valenzuela W et al. 2022. The LUMIERE dataset: Longitudinal glioblastoma MRI with expert RANO evaluation. Sci Data. 9(1):768. https://doi.org/10.1038/s41597-022-01881-7

Taal W, Brandsma D, de Bruin HG et al. 2008. Incidence of early pseudo-progression in a cohort of malignant glioma patients treated with chemoirradiation with temozolomide. Cancer. 113(2):405–410. https://doi.org/10.1002/cncr.23562

Tarek A et al. 2026. TRACE: A concept bottleneck model for longitudinal 3D glioblastoma response assessment. [Preprint, arXiv:2606.30313]. https://arxiv.org/abs/2606.30313

Valerio AG, Trufanova K, de Benedictis S, Vessio G, Castellano G. 2025. From segmentation to explanation: Generating textual reports from MRI with LLMs. Comput Methods Programs Biomed. 270:108922. https://doi.org/10.1016/j.cmpb.2025.108922

Varoquaux G. 2018. Cross-validation failure: Small sample sizes lead to large error bars. NeuroImage. 180:68–77. https://doi.org/10.1016/j.neuroimage.2017.06.061

Wen PY, van den Bent M, Youssef G et al. 2023. RANO 2.0: Update to the Response Assessment in Neuro-Oncology criteria for high- and low-grade gliomas in adults. J Clin Oncol. 41(33):5187–5199.

388 https://doi.org/10.1200/JCO.23.01059
389 Worster A, Carpenter C. 2008. Incorporation bias in studies of diagnostic tests: how to avoid being biased
390 about bias. CJEM. 10(2):174–175. https://doi.org/10.1017/s1481803500009891
391 Zeba M, Al Mamun A, Tithee KJ, Sutradhar D et al. 2026. Mitigating hallucinations in healthcare LLMs with
392 granular fact-checking and domain-specific adaptation. Expert Syst Appl. 329. Preprint arXiv:2512.16189.